

Light Plethysmograph or Photoplethysmography (PPG) is a non-invasive, inexpensive and convenient diagnostic tool used to estimate the skin blood flow using infrared light. It measures oxygen saturation, blood pressure, cardiac output and is used for assessing autonomic functions. The word plethysmograph is a combination of two ancient Greek words ‘plethysmos’ which means increase] and ‘graph’ which is the word for write, and is an instrument mainly used to determine and register the variations in blood volume or blood flow in the body which occur with each heartbeat. It uses a probe which contains a light source and a detector to detect cardio-vascular pulse waves that propagate through the body. The PPG signal reflects the blood movement in the vessel, which goes from the heart to the fingertips and toes through the blood vessels in a wave-like motion. It is an optical measurement technique that uses an invisible infrared light sent into the tissue and the amount of the backscattered light corresponds with the variation of the blood volume.

The diagram illustrates a two-stage circuit. The first stage is a photodiode current-to-voltage converter, consisting of a photodiode (labeled 'photodiode' with an arrow pointing to it) connected to the inverting input of an operational amplifier. The photodiode's cathode is connected to $+V_{CC}$ and its anode to the op-amp input, with current I_p flowing into the input. The op-amp's non-inverting input is grounded ($0V$). A feedback resistor R_{FB} connects the op-amp's output to its inverting input, with feedback current I_{FB} indicated. The output of this stage is V_{CP} . The second stage is a signal conditioner, which takes V_{CP} as input through a resistor C and a parallel combination of a resistor R_1 and a capacitor C . The op-amp in this stage has its non-inverting input grounded and its output V_{out} connected back to its inverting input through a resistor R_2 . The feedback current in this stage is labeled I_{FB} . The circuit is powered by $+V_{CC}$ and $-V_{CC}$ rails.

1. Photodiode Detector (Optical-to-Electrical Conversion)
It converts the received light into a photocurrent. When the LED illuminates the tissue, the detected optical power at the photodiode changes with the cardiac cycle. The photodiode generates a photocurrent I_p approximately proportional to incident light intensity.
2. Current-to-Voltage Converter (Transimpedance Amplifier)

The photodiode produces a current, but most measurement systems (oscilloscopes/ADCs) process voltage. Therefore, the first stage is a transimpedance amplifier (TIA) that converts photocurrent into voltage. The op-amp uses negative feedback through R_{FB} to maintain the inverting input node at a virtual reference (close to 0 V). This keeps the photodiode working in a stable, linear condition and ensures the photodiode current flows through the feedback resistor rather than into the op-amp input.

3. AC Coupling / DC Removal (High-Pass Filtering)

The TIA output V_{op} contains both a large DC offset and a small AC pulsatile waveform. If this mixed signal is amplified directly, the DC level may drive the next amplifier into saturation. To prevent this, the circuit uses a coupling capacitor C and resistor R to form a high-pass filter.

4. Voltage Amplifier Stage (Gain Setting)

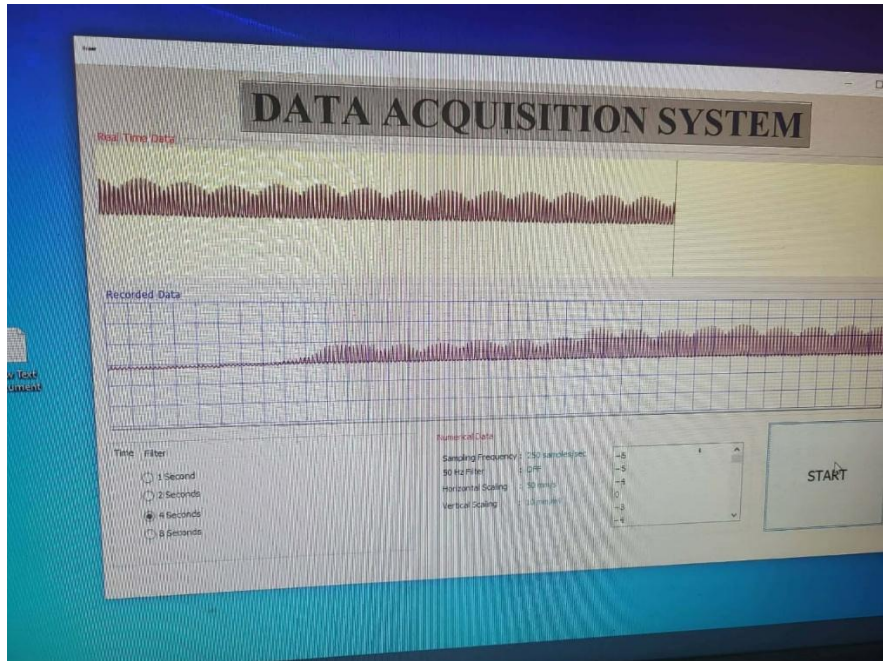
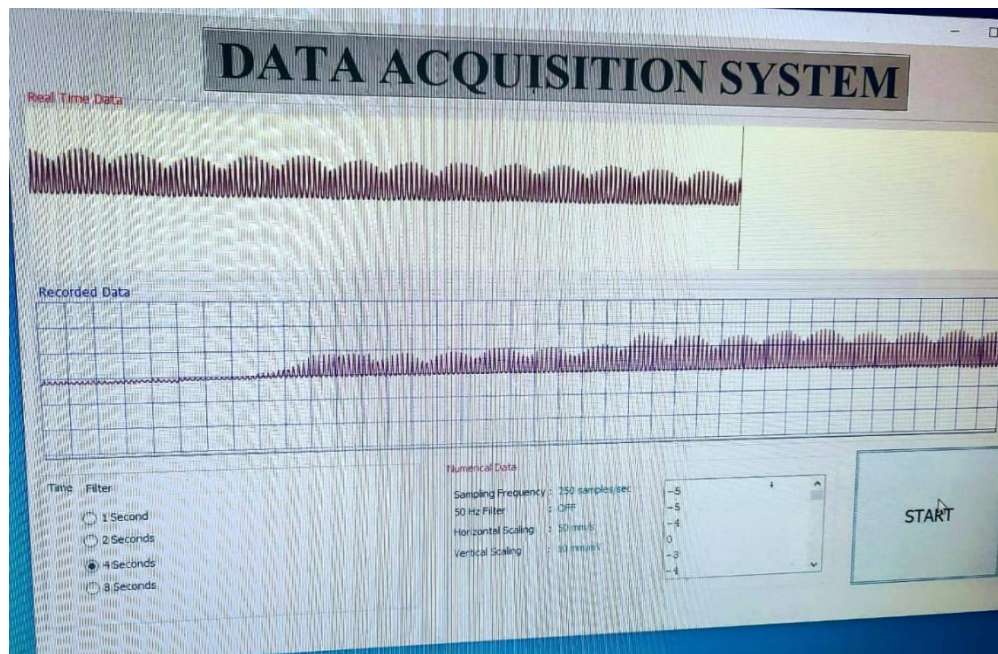
After DC removal, the remaining AC signal is still relatively small. The second op-amp stage amplifies this pulse waveform to a level suitable for observation and recording.

5. Output to Storage Oscilloscope / Data Acquisition

The final conditioned signal is applied to an oscilloscope or recording system. At this point, the waveform primarily represents pulsatile blood volume variation. The waveform can be used to:

- determine heart rate (peak-to-peak interval),
- analyze pulse morphology (rise time, notch, etc.),
- evaluate signal quality (noise/motion artifacts).

Output



Discussion:

The data acquisition system was repeatedly showing straight lines or irregular curves or sometimes no result at all. Possible reasons for this include light interference and hardware malfunction. There is a solar panel beside the LED/Photodetector, it has contacts on both sides. Due to constant use, the contact might have loosened, so it could not show exact pulse rate curve. Also finger movement changes the optical coupling and pressure, producing large low-frequency disturbances often bigger than the pulse itself. Poor ground layout creates ground loops or shared impedance paths that inject hum into the signal. To get real information from the noisy signal, various steps can be taken:

- i) Typical heart rate lies roughly between 0.8–3 Hz Noise outside this range can be reduced using filtering.
- ii) The pulse peaks recur regularly. Even if the waveform is noisy, the largest periodic peaks can often still be identified to estimate heart rate.
- iii) In noisy signals, peak detection works better when thresholds are based on recent signal statistics rather than a fixed voltage level.

To enhance the signal quality, the solar panel contacts should be checked first and if it has any loose contact, that should be fixed at the earliest. Adding optical shielding and blocking of room light can be a solution to light interference. Addition of a low-pass filter after amplification (e.g., cutoff ~ 10 Hz) to remove high-frequency noise while preserving pulse shape is also another way to reduce noise. A notch filter of 50/60 Hz can suppress mains interference if it is prominent.